University of Management and Technology Lahore

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Department of Computer Science

Complex Computing Problem (CCP)

IP Address Optimization and Subnet Management for Large-Scale Institutional Networks

Course: Computer Networks Lab

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1. Problem Statement

This project is based on a real-world network engineering scenario. You have been appointed as a Network Analyst at NetConnect Co., a firm specialising in network architecture and configuration. Your assignment is to design, configure, and simulate a comprehensive network for four companies — JTECH, UCOM, TNet, and NSys — ensuring that every computer inside every company can communicate internally and also reach computers in the other three companies.

Think of it like four large office buildings in different parts of a city. Each building has multiple rooms with computers. We need to give every computer a unique address (IP address), connect them through switches and routers, and make sure any computer in any room of any building can "talk" to a computer in any other building. This is exactly what this project achieves.

Company Requirements at a Glance

Company	Rooms / Depts	PCs per Room	Total PCs	Assigned IP Block
JTECH	5 Rooms	1 PC each	5 PCs	100.186.96.0/19
UCOM	3 Rooms	3 PCs each	9 PCs	35.152.0.0/15
TNet	2 Rooms	4 PCs each	8 PCs	200.98.169.64/26
NSys	4 Depts (2 branches × 2)	DHCP + Static	8 PCs	10.10.0.0/16

NSys is unique. It has two geographically separate branches (like offices in two different cities). Each branch has two departments: Department X (yellow PCs) whose computers receive IP addresses automatically using DHCP, and Department Y (purple PCs) whose computers have manually assigned fixed IP addresses. NSys also operates two dedicated servers — a DHCP & HTTP server and a DNS server.

Key Design Rules

- **One Subnet per Room:** Every room must reside on its own isolated subnet. Devices in different rooms cannot share an IP range.
- **No IP Cross-Sharing:** Each company uses only its assigned IP block. No overlap is permitted between companies.
- **Efficient Subnetting:** CIDR/VLSM (Variable Length Subnet Masking) is used to minimise IP address waste while supporting future growth.
- **Internal Routing — RIP:** The Routing Information Protocol (RIP v2) handles routing within NSys, between its two branches.
- **External Routing — BGP:** The Border Gateway Protocol (BGP) handles routing between all four companies. Each company is a separate Autonomous System (AS).
- **Simulation Requirement:** The entire network must be built in Cisco Packet Tracer and all connectivity must be verified with successful ping tests.

2. Network Architecture Overview

The overall topology follows a hub-and-spoke design. NSys-Branch-1 acts as the central hub for external (inter-company) BGP connections, while the three other companies (JTECH, UCOM, TNet) connect to it via dedicated serial WAN links. NSys itself has an additional serial link between its two branches, over which RIP routing operates.

Each company runs its own internal network using a single router and one or more switches. Subnets (one per room) are configured on the router, and PCs connect to those subnets via switches. WAN connections between companies use /30 point-to-point subnets — the smallest subnet possible, giving only 2 usable IPs (one for each router end) to avoid any address waste.

Topology Diagram

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┌──────────────────────────────────────────────────────────────┐
│                  JTECH  (AS 100)                              │
│  Room1=100.186.96.0/28  Room2=.16/28  Room3=.32/28            │
│  Room4=.48/28          Room5=.64/28                           │
└──────────────────────────────┬───────────────────────────────┘
                               │ Serial WAN  10.0.0.0/30
┌──────────────────────────────┴───────────────────────────────┐
│                              │                                │
│  ┌──────────────┐   Serial WAN   ┌──────────────────────────┐ │
│  │ UCOM AS200   │ 10.0.0.4/30 │   R-NSys-B1  (AS 400)    │ │
│  │ R1=35.152.0  ├─────────────┤ BGP Border + RIP Hub     │ │
│  │  .0/28       │             │ Gi0/0 → B1 Dept X  10.10.1/24 │ │
│  │ R2=.16/28    │             │ Gi0/1 → B1 Dept Y  10.10.2/24 │ │
│  │ R3=.32/28    │ Serial WAN  │ Gi0/2 → Servers    10.10.5/24 │ │
│  └──────────────┘ 10.0.0.8/30 └────────────┬─────────────┘ │
│                                  │ Serial 10.0.0.12/30 │      │
│  ┌──────────────┐      ┌─────────────┴──────────────┐        │
│  │ TNet AS300   ├──────┤   R-NSys-B2  (AS 400)      │        │
│  │ R1=200.98.   │      │ Gi0/0 → B2 Dept X 10.10.3/24 │        │
│  │ 169.64/27    │      │ Gi0/1 → B2 Dept Y 10.10.4/24 │        │
│  │ R2=.96/27    │      └────────────────────────────┘        │
│  └──────────────┘                   │                         │
└──────────────────────────────────────────────────────────────┘
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3. Equipment Used

All devices used in this project are standard components available in Cisco Packet Tracer. The following tables list every piece of equipment, its purpose, and the quantity required.

3.1 Routers

Router Name	Model	Purpose	Serial Module Added
R-JTECH	Cisco 2911	JTECH company router — connects 5 room subnets via VLANs + WAN	HWIC-2T ×1
R-UCOM	Cisco 2911	UCOM company router — connects 3 room subnets + WAN to NSys	HWIC-2T ×1
R-TNET	Cisco 2911	TNet company router — connects 2 room subnets + WAN to NSys	HWIC-2T ×1
R-NSys-B1	Cisco 2911	NSys Branch 1 — BGP border, RIP hub, DHCP relay, server gateway	HWIC-2T ×2
R-NSys-B2	Cisco 2911	NSys Branch 2 — connects B2 departments, RIP to B1	HWIC-2T ×1

Total Routers: 5 | Each HWIC-2T module adds 2 serial WAN ports.

3.2 Switches

Switch Name	Model	Connected To	Configuration
SW-JTECH	2960	All 5 JTECH PCs + R-JTECH Gi0/0	VLANs 10–50 + Trunk port
SW-UCOM-R1	2960	3 UCOM Room 1 PCs + R-UCOM Gi0/0	Default (no VLAN config)
SW-UCOM-R2	2960	3 UCOM Room 2 PCs + R-UCOM Gi0/1	Default
SW-UCOM-R3	2960	3 UCOM Room 3 PCs + R-UCOM Gi0/2	Default
SW-TNET-R1	2960	4 TNet Room 1 PCs + R-TNET Gi0/0	Default
SW-TNET-R2	2960	4 TNet Room 2 PCs + R-TNET Gi0/1	Default
SW-NSys-B1-DeptX	2960	B1 Dept X PCs + R-NSys-B1 Gi0/0	Default
SW-NSys-B1-DeptY	2960	B1 Dept Y PCs + R-NSys-B1 Gi0/1	Default
SW-NSys-Servers	2960	Both servers + R-NSys-B1 Gi0/2	Default
SW-NSys-B2-DeptX	2960	B2 Dept X PCs + R-NSys-B2 Gi0/0	Default
SW-NSys-B2-DeptY	2960	B2 Dept Y PCs + R-NSys-B2	Default

		Gi0/1	

Total Switches: 11

3.3 End Devices & Servers

Device	Qty	Location	IP Assignment Method
PC (Generic)	5	JTECH — 1 per room	Static (manual)
PC (Generic)	9	UCOM — 3 per room × 3 rooms	Static (manual)
PC (Generic)	8	TNet — 4 per room × 2 rooms	Static (manual)
PC — Yellow	4	NSys Dept X — 2 per branch	DHCP (automatic)
PC — Purple	4	NSys Dept Y — 2 per branch	Static (manual)
Server (Generic)	1	NSys Server Subnet (10.10.5.10)	Static — runs DHCP & HTTP
Server (Generic)	1	NSys Server Subnet (10.10.5.11)	Static — runs DNS

Total PCs: 30 | Total Servers: 2

3.4 Cables

Cable Type	Used For	Approx. Count
Copper Straight-Through	PC → Switch, Server → Switch, Switch → Router LAN port	~33
Serial DCE	Router → Router WAN links (inter-company + inter-branch)	4

NOTE: When connecting Serial DCE cables in Packet Tracer, always click the R-NSys-B1 router FIRST. Whichever device you click first becomes the DCE (clock) end and must set the clock rate command.

4. Phase 1 — Requirements Analysis

Before configuring any device, we must fully understand the network requirements. This phase documents the terminology, key concepts, and device count so any reader — regardless of prior networking experience — can follow the project.

Key Concepts Explained

What is an IP Address?

An IP (Internet Protocol) address is a unique numerical label assigned to every device on a network — like a home postal address, but for computers. For example, 100.186.96.2 is an IP address. IPv4 addresses are 32 bits long, written as four groups of numbers separated by dots, each ranging from 0 to 255.

What is a Subnet?

A subnet (sub-network) is a smaller network carved out of a larger IP block. Devices in the same subnet can communicate directly. To reach a different subnet, data must pass through a router. In this project, each room is its own subnet so that traffic is isolated and managed per room.

What is CIDR / VLSM?

CIDR (Classless Inter-Domain Routing) uses prefix notation like /28 to describe how many bits are the "network" part of an address. VLSM (Variable Length Subnet Mask) allows different subnets to have different sizes. We use /28 for small rooms (14 usable hosts), /27 for medium rooms (30 hosts), and /30 for WAN links (only 2 hosts needed per link).

What is a Default Gateway?

Every PC must know its default gateway — the IP address of the router interface in its subnet. When a PC wants to send data outside its subnet (to a different room or a different company), it sends the packet to the gateway, and the router handles the rest.

Device Count Summary

Company	Rooms	PCs	Router	Switches	WAN Connection
JTECH	5	5	1 (R-JTECH)	1 (with VLANs)	Serial → NSys-B1
UCOM	3	9	1 (R-UCOM)	3 (one/room)	Serial → NSys-B1
TNet	2	8	1 (R-TNET)	2 (one/room)	Serial → NSys-B1
NSys B1	2 Depts	4+2	1 (R-NSys-B1)	3 switches	BGP hub for all
NSys B2	2 Depts	4	1 (R-NSys-B2)	2 switches	Serial → NSys-B1 (RIP)

5. Phase 2 — Subnetting & IP Address Planning

Subnetting is the process of dividing a large IP block into smaller sub-networks. We use VLSM to choose the right subnet size for each room — large enough for current needs plus future growth, but small enough to avoid wasting addresses. The subnet size is determined by the number of hosts (devices) needed.

Subnet Size Reference Table

Prefix	Subnet Mask	Total IPs	Usable Hosts	Best Used For
/30	255.255.255.252	4	2	Router-to-router WAN links (point-to-point)
/29	255.255.255.248	8	6	Very small subnets — up to 6 devices
/28	255.255.255.240	16	14	Small room subnets — 1 to 14 devices
/27	255.255.255.224	32	30	Medium subnets — up to 30 devices
/24	255.255.255.0	256	254	Standard LAN — up to 254 devices
/15	255.254.0.0	131,072	131,070	Very large corporate blocks
/19	255.255.224.0	8,192	8,190	Large departmental blocks

5.1 JTECH Subnetting — 100.186.96.0/19

JTECH is assigned the block 100.186.96.0/19, giving 2^{13} = 8,192 total IP addresses (100.186.96.0 through 100.186.127.255). JTECH has 5 rooms with 1 PC each. We choose /28 per room (14 usable hosts) to allow up to 13 more PCs per room in future. Subnetting starts from the beginning of the block; each /28 takes 16 consecutive addresses.

Room	Subnet ID	Subnet Mask	Usable IP Range	Broadcast Address	Gateway (Router)	Max Hosts
Room 1	100.186.96.0	255.255.255.240 /28	100.186.96.1 – .14	100.186.96.15	100.186.96.1	14
Room 2	100.186.96.16	255.255.255.240 /28	100.186.96.17 – .30	100.186.96.31	100.186.96.17	14
Room 3	100.186.96.32	255.255.255.240 /28	100.186.96.33 – .46	100.186.96.47	100.186.96.33	14
Room 4	100.186.96.48	255.255.255.240 /28	100.186.96.49 – .62	100.186.96.63	100.186.96.49	14
Room 5	100.186.96.64	255.255.255.240 /28	100.186.96.65 – .78	100.186.96.79	100.186.96.65	14

NOTE: IPs used: 80 of 8,192. Remaining: 8,112 (500+ future /28 subnets possible).

5.2 UCOM Subnetting — 35.152.0.0/15

UCOM is assigned 35.152.0.0/15, giving 2^{17} = 131,072 total IP addresses (35.152.0.0 through 35.153.255.255). UCOM has 3 rooms with 3 PCs each. We use /28 per room (14 hosts) — satisfies current need of 3 PCs and supports 11 more per room.

Room	Subnet ID	Subnet Mask	Usable IP Range	Broadcast Address	Gateway (Router)	Max Hosts
Room 1	35.152.0.0	255.255.255.240 /28	35.152.0.1 – .14	35.152.0.15	35.152.0.1	14
Room 2	35.152.0.16	255.255.255.240 /28	35.152.0.17 – .30	35.152.0.31	35.152.0.17	14
Room 3	35.152.0.32	255.255.255.240 /28	35.152.0.33 – .46	35.152.0.47	35.152.0.33	14

NOTE: IPs used: 48 of 131,072. Utilisation < 0.04%. Enormous future expansion capacity.

5.3 TNet Subnetting — 200.98.169.64/26

TNet is assigned 200.98.169.64/26, giving 2^6 = 64 total IP addresses (200.98.169.64 through 200.98.169.127). TNet has 2 rooms with 4 PCs each. We split the /26 exactly in half into 2 × /27 subnets (30 usable hosts each). This achieves 100% block utilisation with no wasted addresses, and each room can grow to 30 PCs.

Room	Subnet ID	Subnet Mask	Usable IP Range	Broadcast Address	Gateway (Router)	Max Hosts
Room 1	200.98.169.64	255.255.255.224 /27	200.98.169.65 – .94	200.98.169.95	200.98.169.65	30
Room 2	200.98.169.96	255.255.255.224 /27	200.98.169.97 – .126	200.98.169.127	200.98.169.97	30

NOTE: Entire /26 block used: 2 subnets × 32 = 64 addresses. Perfect utilisation. 26 spare hosts per room.

5.4 NSys Subnetting — 10.10.0.0/16

NSys is assigned 10.10.0.0/16 (65,536 addresses). Since NSys has two branches, multiple departments, and centralised servers, we use /24 subnets (254 usable hosts each). /24 is standard for department LANs and provides ample room for many future devices.

Location	Subnet	Mask	Usable Range	Broadcast	Gateway	IP Method
Branch 1 — Dept X	10.10.1.0	/24 255.255.255.0	10.10.1.1 – .254	10.10.1.255	10.10.1.1	DHCP (Yellow PCs)
Branch 1 — Dept Y	10.10.2.0	/24 255.255.255.0	10.10.2.1 – .254	10.10.2.255	10.10.2.1	Static (Purple PCs)
Branch 2 — Dept X	10.10.3.0	/24 255.255.255.0	10.10.3.1 – .254	10.10.3.255	10.10.3.1	DHCP (Yellow

						PCs)
Branch 2 — Dept Y	10.10.4.0	/24 255.255.255.0	10.10.4.1 – .254	10.10.4.255	10.10.4.1	Static (Purple PCs)
Server Subnet	10.10.5.0	/24 255.255.255.0	10.10.5.1 – .254	10.10.5.255	10.10.5.1	Static Servers

5.5 WAN Serial Link Addressing

Serial WAN links between routers use /30 subnets — the smallest subnet in networking, providing exactly 2 usable IPs (one per router end). The DCE end (always R-NSys-B1 in this project) must set the clock rate to synchronise the serial connection speed.

WAN Link	Subnet	NSys-B1 IP (DCE)	Other Router IP (DTE)	NSys-B1 Port	Other Port
NSys-B1 ↔ JTECH	10.0.0.0/30	10.0.0.2	10.0.0.1	Se0/0/0	Se0/0/0
NSys-B1 ↔ UCOM	10.0.0.4/30	10.0.0.6	10.0.0.5	Se0/0/1	Se0/0/0
NSys-B1 ↔ TNet	10.0.0.8/30	10.0.0.10	10.0.0.9	Se0/1/0	Se0/0/0
NSys-B1 ↔ NSys-B2	10.0.0.12/30	10.0.0.13	10.0.0.14	Se0/1/1	Se0/0/0

6. Complete PC IP Configuration Table

Every static PC must be configured manually with an IP address, subnet mask, and default gateway. In Packet Tracer: click a PC → Desktop tab → IP Configuration → enter values. For DHCP PCs (NSys Dept X), simply select the "DHCP" radio button and the IP is assigned automatically.

JTECH PCs

PC Name	IP Address	Subnet Mask	Default Gateway
JTECH-R1-PC1	100.186.96.2	255.255.255.240	100.186.96.1
JTECH-R2-PC1	100.186.96.18	255.255.255.240	100.186.96.17
JTECH-R3-PC1	100.186.96.34	255.255.255.240	100.186.96.33
JTECH-R4-PC1	100.186.96.50	255.255.255.240	100.186.96.49
JTECH-R5-PC1	100.186.96.66	255.255.255.240	100.186.96.65

UCOM PCs

PC Name	IP Address	Subnet Mask	Default Gateway
UCOM-R1-PC1	35.152.0.2	255.255.255.240	35.152.0.1
UCOM-R1-PC2	35.152.0.3	255.255.255.240	35.152.0.1
UCOM-R1-PC3	35.152.0.4	255.255.255.240	35.152.0.1
UCOM-R2-PC1	35.152.0.18	255.255.255.240	35.152.0.17
UCOM-R2-PC2	35.152.0.19	255.255.255.240	35.152.0.17
UCOM-R2-PC3	35.152.0.20	255.255.255.240	35.152.0.17
UCOM-R3-PC1	35.152.0.34	255.255.255.240	35.152.0.33
UCOM-R3-PC2	35.152.0.35	255.255.255.240	35.152.0.33
UCOM-R3-PC3	35.152.0.36	255.255.255.240	35.152.0.33

TNet PCs

PC Name	IP Address	Subnet Mask	Default Gateway
TNET-R1-PC1	200.98.169.66	255.255.255.224	200.98.169.65
TNET-R1-PC2	200.98.169.67	255.255.255.224	200.98.169.65
TNET-R1-PC3	200.98.169.68	255.255.255.224	200.98.169.65
TNET-R1-PC4	200.98.169.69	255.255.255.224	200.98.169.65
TNET-R2-PC1	200.98.169.98	255.255.255.224	200.98.169.97
TNET-R2-PC2	200.98.169.99	255.255.255.224	200.98.169.97
TNET-R2-PC3	200.98.169.100	255.255.255.224	200.98.169.97
TNET-R2-PC4	200.98.169.101	255.255.255.224	200.98.169.97

NSys PCs & Servers

Device Name	IP Address	Subnet Mask	Default Gateway	Method
NSys-B1-DeptX-PC1	Auto via DHCP (~10.10.1.10)	255.255.255.0	10.10.1.1	DHCP
NSys-B1-DeptX-PC2	Auto via DHCP (~10.10.1.11)	255.255.255.0	10.10.1.1	DHCP
NSys-B1-DeptY-PC1	10.10.2.10	255.255.255.0	10.10.2.1	Static
NSys-B1-DeptY-PC2	10.10.2.11	255.255.255.0	10.10.2.1	Static
NSys-B2-DeptX-PC1	Auto via DHCP (~10.10.3.10)	255.255.255.0	10.10.3.1	DHCP
NSys-B2-DeptX-PC2	Auto via DHCP (~10.10.3.11)	255.255.255.0	10.10.3.1	DHCP
NSys-B2-DeptY-PC1	10.10.4.10	255.255.255.0	10.10.4.1	Static
NSys-B2-DeptY-PC2	10.10.4.11	255.255.255.0	10.10.4.1	Static
SVR-DHCP-HTTP	10.10.5.10	255.255.255.0	10.10.5.1	Static
SVR-DNS	10.10.5.11	255.255.255.0	10.10.5.1	Static

7. Phase 3 — Packet Tracer Build Guide (Step-by-Step)

This section provides a complete, step-by-step walkthrough for building the network topology in Cisco Packet Tracer. Follow each step in order. If you are new to Packet Tracer, every device used is in the default device library — no extra downloads are needed.

Step 1 — Open Packet Tracer and Place Devices

From the bottom-left device panel in Packet Tracer:

1. Select "Routers" → choose model "2911" → click 5 times on the canvas to place R-JTECH, R-UCOM, R-TNET, R-NSys-B1, R-NSys-B2. Label each one immediately.
2. Select "Switches" → choose model "2960" → place 11 switches (one per name listed in Section 3.2).
3. Select "End Devices" → choose "PC" → place 30 PCs. Colour the NSys Dept X PCs yellow and Dept Y PCs purple using the device colour option.
4. Select "End Devices" → choose "Server" → place 2 servers near the NSys area. Label them SVR-DHCP-HTTP and SVR-DNS.
5. Arrange devices in logical groups on the canvas — JTECH top-left, UCOM top-right, TNet bottom-left, NSys bottom-right.

Step 2 — Add Serial Modules to Each Router

Routers require serial port modules for WAN links. The Cisco 2911 has no serial ports by default. Repeat these steps for every router:

1. Click on the router to open its configuration window.
2. Click the "Physical" tab at the top of the window.
3. Click the green POWER button to turn the router OFF (required before inserting modules).
4. In the left module panel, find "HWIC-2T" and drag it into any empty slot on the router picture.
5. For R-NSys-B1 ONLY: add TWO "HWIC-2T" modules — one in slot 0, one in slot 1. It needs 4 serial ports.
6. Click the power button again to turn the router back ON.

Step 3 — Connect All Cables

Straight-Through Copper Cables (PC/Server → Switch, Switch → Router)

In Packet Tracer: click the orange lightning bolt (Connections) in the bottom panel → select "Copper Straight-Through". Click the source device, choose a port, then click the destination.

From	From Port	To	To Port
JTECH-R1-PC1	Fa0	SW-JTECH	Fa0/2
JTECH-R2-PC1	Fa0	SW-JTECH	Fa0/3
JTECH-R3-PC1	Fa0	SW-JTECH	Fa0/4
JTECH-R4-PC1	Fa0	SW-JTECH	Fa0/5
JTECH-R5-PC1	Fa0	SW-JTECH	Fa0/6
SW-JTECH	Fa0/1	R-JTECH	Gi0/0 (trunk)
UCOM-R1-PC1/2/3	Fa0	SW-UCOM-R1	Fa0/2, Fa0/3, Fa0/4
UCOM-R2-PC1/2/3	Fa0	SW-UCOM-R2	Fa0/2, Fa0/3, Fa0/4
UCOM-R3-PC1/2/3	Fa0	SW-UCOM-R3	Fa0/2, Fa0/3, Fa0/4
SW-UCOM-R1	Fa0/1	R-UCOM	Gi0/0
SW-UCOM-R2	Fa0/1	R-UCOM	Gi0/1
SW-UCOM-R3	Fa0/1	R-UCOM	Gi0/2
TNET-R1-PC1/2/3/4	Fa0	SW-TNET-R1	Fa0/2 – Fa0/5
TNET-R2-PC1/2/3/4	Fa0	SW-TNET-R2	Fa0/2 – Fa0/5
SW-TNET-R1	Fa0/1	R-TNET	Gi0/0
SW-TNET-R2	Fa0/1	R-TNET	Gi0/1
NSys-B1 DeptX PCs	Fa0	SW-NSys-B1-DeptX	Fa0/2, Fa0/3
SW-NSys-B1-DeptX	Fa0/1	R-NSys-B1	Gi0/0
NSys-B1 DeptY PCs	Fa0	SW-NSys-B1-DeptY	Fa0/2, Fa0/3
SW-NSys-B1-DeptY	Fa0/1	R-NSys-B1	Gi0/1
SVR-DHCP-HTTP	Fa0	SW-NSys-Servers	Fa0/2
SVR-DNS	Fa0	SW-NSys-Servers	Fa0/3
SW-NSys-Servers	Fa0/1	R-NSys-B1	Gi0/2
NSys-B2 DeptX PCs	Fa0	SW-NSys-B2-DeptX	Fa0/2, Fa0/3
SW-NSys-B2-DeptX	Fa0/1	R-NSys-B2	Gi0/0
NSys-B2 DeptY PCs	Fa0	SW-NSys-B2-DeptY	Fa0/2, Fa0/3
SW-NSys-B2-DeptY	Fa0/1	R-NSys-B2	Gi0/1

Serial DCE Cables (Router ↔ Router WAN Links)

Select "Serial DCE" cable from the connections panel. IMPORTANT: Click the DCE router (R-NSys-B1) first in every case.

Click 1st (DCE) → R-NSys-B1	DCE Port	Click 2nd (DTE) Router	DTE Port
R-NSys-B1	Se0/0/0	R-JTECH	Se0/0/0
R-NSys-B1	Se0/0/1	R-UCOM	Se0/0/0
R-NSys-B1	Se0/1/0	R-TNET	Se0/0/0
R-NSys-B1	Se0/1/1	R-NSys-B2	Se0/0/0

8. Phase 4 — Router & Switch CLI Configurations

The CLI (Command Line Interface) is the text console used to configure routers and switches. In Packet Tracer, click a device → click the "CLI" tab → type "enable" to enter privileged mode. All commands below can be typed or copy-pasted. Lines starting with "!" are green comments that explain each command — the router ignores them.

8.1 R-JTECH — JTECH Company Router

R-JTECH uses a technique called Router-on-a-Stick. Because JTECH has 5 rooms but the 2911 router only has 3 physical GigabitEthernet ports, we create 5 logical sub-interfaces on one physical port (Gi0/0). Each sub-interface handles one room's subnet, tagged with an 802.1Q VLAN number. The switch must have matching VLAN configuration (see Section 8.2).

```
enable
configure terminal
hostname R-JTECH

! The physical port must be activated — sub-interfaces inherit this state
interface GigabitEthernet0/0
 no shutdown

! Sub-interface for Room 1 — VLAN tag 10 — Subnet 100.186.96.0/28
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 100.186.96.1 255.255.255.240
 no shutdown

! Sub-interface for Room 2 — VLAN tag 20 — Subnet 100.186.96.16/28
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 100.186.96.17 255.255.255.240
 no shutdown

! Sub-interface for Room 3 — VLAN tag 30 — Subnet 100.186.96.32/28
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 100.186.96.33 255.255.255.240
 no shutdown

! Sub-interface for Room 4 — VLAN tag 40 — Subnet 100.186.96.48/28
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 100.186.96.49 255.255.255.240
 no shutdown

! Sub-interface for Room 5 — VLAN tag 50 — Subnet 100.186.96.64/28
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 100.186.96.65 255.255.255.240
 no shutdown
```

```
! WAN serial link to NSys-B1 — JTECH is the DTE end (no clock rate here)
interface Serial0/0/0
 ip address 10.0.0.1 255.255.255.252
 no shutdown

! RIP v2 — advertises JTECH subnets internally (single router so minimal effect here)
router rip
 version 2
 network 100.0.0.0
 no auto-summary

! BGP — JTECH is Autonomous System 100
! We tell BGP about our BGP neighbour at NSys-B1 (AS 400) using its serial IP
! The 'network' command advertises the entire JTECH /19 block to all BGP peers
router bgp 100
 bgp log-neighbor-changes
 neighbor 10.0.0.2 remote-as 400
 network 100.186.96.0 mask 255.255.224.0
 no auto-summary

end
write memory
```

8.2 SW-JTECH — JTECH Central Switch (VLAN Configuration)

This switch must be configured with 5 VLANs (one per room) to match the router sub-interfaces. Port Fa0/1 is the trunk port connecting to the router — it carries all VLAN traffic with tags. Ports Fa0/2 through Fa0/6 are access ports, each belonging to exactly one VLAN (one room). PCs plug into these access ports and have no awareness of VLANs.

```
enable
configure terminal
hostname SW-JTECH

! Create the 5 VLANs — one per JTECH room
vlan 10
 name JTECH-Room1
vlan 20
 name JTECH-Room2
vlan 30
 name JTECH-Room3
vlan 40
 name JTECH-Room4
vlan 50
 name JTECH-Room5
exit

! Trunk port — connects to R-JTECH Gi0/0 — carries all VLAN traffic tagged
interface FastEthernet0/1
 switchport mode trunk
```

```
 no shutdown

! Room 1 access port — PC in Room 1 plugs here (VLAN 10 only)
interface FastEthernet0/2
 switchport mode access
 switchport access vlan 10
 no shutdown

! Room 2 access port — VLAN 20 only
interface FastEthernet0/3
 switchport mode access
 switchport access vlan 20
 no shutdown

! Room 3 access port — VLAN 30 only
interface FastEthernet0/4
 switchport mode access
 switchport access vlan 30
 no shutdown

! Room 4 access port — VLAN 40 only
interface FastEthernet0/5
 switchport mode access
 switchport access vlan 40
 no shutdown

! Room 5 access port — VLAN 50 only
interface FastEthernet0/6
 switchport mode access
 switchport access vlan 50
 no shutdown

end
write memory
```

8.3 R-UCOM — UCOM Company Router

UCOM has only 3 rooms, so we can use 3 separate GigabitEthernet ports — one per room subnet. This is simpler than VLANs. Each port connects to its own switch (SW-UCOM-R1, R2, R3). Those switches require NO configuration — leave them at factory defaults.

```
enable
configure terminal
hostname R-UCOM

! Room 1 — GE0/0 — Subnet 35.152.0.0/28
interface GigabitEthernet0/0
 ip address 35.152.0.1 255.255.255.240
 no shutdown

! Room 2 — GE0/1 — Subnet 35.152.0.16/28
```

```
interface GigabitEthernet0/1
 ip address 35.152.0.17 255.255.255.240
 no shutdown

! Room 3 — GE0/2 — Subnet 35.152.0.32/28
interface GigabitEthernet0/2
 ip address 35.152.0.33 255.255.255.240
 no shutdown

! WAN serial link to NSys-B1 — UCOM is the DTE end
interface Serial0/0/0
 ip address 10.0.0.5 255.255.255.252
 no shutdown

! RIP v2 — 'network 35.0.0.0' matches the Class A range covering UCOM's subnets
router rip
 version 2
 network 35.0.0.0
 no auto-summary

! BGP — UCOM is Autonomous System 200
! Neighbour: NSys-B1 at 10.0.0.6 (AS 400)
! Advertising entire UCOM /15 block
router bgp 200
 bgp log-neighbor-changes
 neighbor 10.0.0.6 remote-as 400
 network 35.152.0.0 mask 255.254.0.0
 no auto-summary

end
write memory
```

8.4 R-TNET — TNet Company Router

TNet has 2 rooms. GE0/0 serves Room 1 and GE0/1 serves Room 2 — no VLANs needed. Each port connects to its own switch (SW-TNET-R1 and SW-TNET-R2), both of which require no configuration.

```
enable
configure terminal
hostname R-TNET

! Room 1 — GE0/0 — Subnet 200.98.169.64/27
interface GigabitEthernet0/0
 ip address 200.98.169.65 255.255.255.224
 no shutdown

! Room 2 — GE0/1 — Subnet 200.98.169.96/27
interface GigabitEthernet0/1
 ip address 200.98.169.97 255.255.255.224
 no shutdown
```

```
! WAN serial link to NSys-B1 — TNet is the DTE end
interface Serial0/0/0
 ip address 10.0.0.9 255.255.255.252
 no shutdown

! RIP v2 — 'network 200.98.0.0' covers TNet's 200.98.x.x subnets
router rip
 version 2
 network 200.98.0.0
 no auto-summary

! BGP — TNet is Autonomous System 300
! Neighbour: NSys-B1 at 10.0.0.10 (AS 400)
! Advertising entire TNet /26 block
router bgp 300
 bgp log-neighbor-changes
 neighbor 10.0.0.10 remote-as 400
 network 200.98.169.64 mask 255.255.255.192
 no auto-summary

end
write memory
```

8.5 R-NSys-B1 — NSys Branch 1 Router (BGP Border + RIP Hub + DHCP Relay)

R-NSys-B1 performs four critical roles simultaneously:

- **BGP Border Router:** Establishes BGP peer sessions with JTECH (AS100), UCOM (AS200), and TNet (AS300). Advertises NSys subnets to all three.
- **RIP Internal Hub:** Exchanges RIP routing updates with R-NSys-B2 so both branches can reach each other.
- **Route Redistributor:** Injects BGP-learned external routes into RIP (so B2 can reach JTECH/UCOM/TNet), and injects RIP-learned B2 routes into BGP (so external companies can reach B2).
- **DHCP Relay Agent:** "ip helper-address" on Gi0/0 intercepts DHCP broadcast requests from Dept X PCs and forwards them as unicast to the DHCP server (10.10.5.10). Without this, DHCP only works on the local subnet.

```
enable
configure terminal
hostname R-NSys-B1

! Branch 1 Dept X — DHCP yellow PCs
! ip helper-address: forwards DHCP broadcasts from this interface to the DHCP server
interface GigabitEthernet0/0
```

```
 ip address 10.10.1.1 255.255.255.0
 ip helper-address 10.10.5.10
 no shutdown

! Branch 1 Dept Y — static purple PCs (no DHCP relay needed)
interface GigabitEthernet0/1
 ip address 10.10.2.1 255.255.255.0
 no shutdown

! Server subnet — hosts SVR-DHCP-HTTP and SVR-DNS
interface GigabitEthernet0/2
 ip address 10.10.5.1 255.255.255.0
 no shutdown

! WAN to JTECH — NSys-B1 is DCE → must set clock rate to sync the link
interface Serial0/0/0
 ip address 10.0.0.2 255.255.255.252
 clock rate 64000
 no shutdown

! WAN to UCOM — DCE end
interface Serial0/0/1
 ip address 10.0.0.6 255.255.255.252
 clock rate 64000
 no shutdown

! WAN to TNet — DCE end
interface Serial0/1/0
 ip address 10.0.0.10 255.255.255.252
 clock rate 64000
 no shutdown

! Inter-branch WAN to NSys-B2 — DCE end
interface Serial0/1/1
 ip address 10.0.0.13 255.255.255.252
 clock rate 64000
 no shutdown

! RIP v2 — internal NSys routing between branches
! 'network 10.0.0.0' enables RIP on all 10.x.x.x interfaces (LAN + serial)
! passive-interface: blocks RIP updates from going out to JTECH/UCOM/TNet WAN ports
!   (they only speak BGP, sending RIP to them causes errors)
! redistribute bgp: routes from JTECH/UCOM/TNet (learned via BGP) are shared
!   with NSys-B2 via RIP so Branch 2 can reach external companies
router rip
 version 2
 network 10.0.0.0
 passive-interface Serial0/0/0
 passive-interface Serial0/0/1
 passive-interface Serial0/1/0
 redistribute bgp 400 metric 5
 no auto-summary
```

```
! BGP — NSys is Autonomous System 400
! no synchronization: allows BGP to advertise routes without waiting for IGP (needed in PT)
! Three external neighbours: JTECH (100), UCOM (200), TNet (300)
! network: advertise NSys B1 subnets directly
! redistribute rip: B2 subnets (learned via RIP) are also advertised to external companies
router bgp 400
 no synchronization
 bgp log-neighbor-changes
 neighbor 10.0.0.1 remote-as 100
 neighbor 10.0.0.5 remote-as 200
 neighbor 10.0.0.9 remote-as 300
 network 10.10.1.0 mask 255.255.255.0
 network 10.10.2.0 mask 255.255.255.0
 network 10.10.5.0 mask 255.255.255.0
 redistribute rip
 no auto-summary

end
write memory
```

8.6 R-NSys-B2 — NSys Branch 2 Router

Branch 2 router is simpler — it only runs RIP. It has no direct BGP connection to external companies. Routes to JTECH, UCOM, and TNet arrive via RIP from R-NSys-B1 (which redistributes BGP routes into RIP). The DHCP relay on Gi0/0 forwards Dept X requests through the serial link to the central DHCP server at 10.10.5.10.

```
enable
configure terminal
hostname R-NSys-B2

! Branch 2 Dept X — DHCP yellow PCs
! ip helper-address relays DHCP requests via serial link to server at 10.10.5.10
interface GigabitEthernet0/0
 ip address 10.10.3.1 255.255.255.0
 ip helper-address 10.10.5.10
 no shutdown

! Branch 2 Dept Y — static purple PCs
interface GigabitEthernet0/1
 ip address 10.10.4.1 255.255.255.0
 no shutdown

! WAN serial link to NSys-B1 — B2 is the DTE end (no clock rate)
interface Serial0/0/0
 ip address 10.0.0.14 255.255.255.252
 no shutdown

! RIP v2 — shares B2 subnets with B1, receives all external routes from B1
router rip
```

```
 version 2
 network 10.0.0.0
 no auto-summary

end
write memory
```

9. Server Configuration (DHCP, HTTP, DNS)

Servers in Packet Tracer are configured through a graphical interface. Click the server → "Desktop" tab for IP settings → "Services" tab for services. No CLI commands are needed for servers.

9.1 SVR-DHCP-HTTP — Static IP Configuration

Click SVR-DHCP-HTTP → Desktop → IP Configuration:

Field	Value to Enter
IP Address	10.10.5.10
Subnet Mask	255.255.255.0
Default Gateway	10.10.5.1
DNS Server	10.10.5.11

9.2 DHCP Service — Two Pools for Both Branches

The DHCP service assigns IP addresses automatically to NSys Dept X PCs in both branches. We create two pools — one for each branch. The router's "ip helper-address" ensures that DHCP requests from both branches reach this server even though they are on different subnets.

Click SVR-DHCP-HTTP → Services → DHCP → toggle to ON → add two pools:

Field	Pool 1 — Branch 1 Dept X	Pool 2 — Branch 2 Dept X
Pool Name	NSys-B1-DeptX	NSys-B2-DeptX
Default Gateway	10.10.1.1	10.10.3.1
DNS Server	10.10.5.11	10.10.5.11
Start IP Address	10.10.1.10	10.10.3.10
Subnet Mask	255.255.255.0	255.255.255.0
Maximum Users	50	50

NOTE: Click "Add" after entering each pool. The yellow (Dept X) PCs must have their IP Configuration set to DHCP mode in Packet Tracer.

9.3 HTTP Service

Click SVR-DHCP-HTTP → Services → HTTP → ensure it is ON. The default webpage is sufficient. Any PC can test HTTP by opening the Web Browser (Desktop tab on a PC) and navigating to http://10.10.5.10.

9.4 SVR-DNS — Static IP Configuration

Click SVR-DNS → Desktop → IP Configuration:

Field	Value to Enter

IP Address	10.10.5.11
Subnet Mask	255.255.255.0
Default Gateway	10.10.5.1

9.5 DNS Service

Click SVR-DNS → Services → DNS → toggle to ON → add these records:

Name (Domain)	Record Type	Address	Purpose
www.nsys.com	A Record	10.10.5.10	Maps domain to the DHCP/HTTP server
jtech.local	A Record	100.186.96.1	Maps name to JTECH router interface

10. Routing Protocol Configuration (RIP & BGP)

10.1 RIP — Routing Information Protocol

RIP is a simple interior gateway protocol used within a single organisation. It works by having each router periodically broadcast its routing table to its neighbours. Neighbours update their tables and re-broadcast. RIP uses "hop count" as its metric — the number of routers a packet must pass through to reach a destination. Maximum hop count is 15; anything beyond is considered unreachable.

In this project, RIP v2 runs only within NSys — between R-NSys-B1 and R-NSys-B2. This allows Branch 1 to know Branch 2's subnets (10.10.3.0/24, 10.10.4.0/24) and vice versa. RIP v2 is used (not v1) because it supports VLSM — it includes subnet masks in its updates.

10.2 BGP — Border Gateway Protocol

BGP is the routing protocol of the internet. It is used between different organisations (Autonomous Systems). Unlike RIP which automatically discovers neighbours, BGP requires explicit configuration of each neighbour's IP address and AS number. BGP is highly controllable — administrators choose exactly which routes to advertise and accept.

Each company has a unique AS number:

Company	AS Number	BGP Role	BGP Peers
JTECH	AS 100	Stub AS — advertises 100.186.96.0/19	NSys-B1 only
UCOM	AS 200	Stub AS — advertises 35.152.0.0/15	NSys-B1 only
TNet	AS 300	Stub AS — advertises 200.98.169.64/26	NSys-B1 only
NSys-B1	AS 400	Transit AS — hub for all inter-company BGP	JTECH, UCOM, TNet

10.3 How RIP and BGP Work Together (Redistribution)

On R-NSys-B1, the two protocols are connected through route redistribution — the process of taking routes learned from one protocol and injecting them into another.

- **redistribute bgp 400 metric 5 (inside RIP config):** Routes learned from external companies via BGP are shared with NSys-B2 via RIP. This allows B2 to reach JTECH, UCOM, and TNet even though B2 doesn't speak BGP.
- **redistribute rip (inside BGP config):** Routes learned from NSys-B2 via RIP (10.10.3.0/24, 10.10.4.0/24) are advertised to external companies via BGP. This allows JTECH, UCOM, TNet to reach B2 subnets.
- **passive-interface Serial0/0/0-1, Serial0/1/0 (in RIP):** Stops RIP update packets from being sent out the WAN interfaces toward JTECH, UCOM, TNet. Those routers only speak BGP and would reject or mishandle RIP updates.

10.4 Packet Route Trace Example

Scenario: JTECH-R1-PC1 (100.186.96.2) sends a ping to NSys-B2-DeptY-PC1 (10.10.4.10):

Step 1: JTECH-R1-PC1 sends the ping packet to its default gateway: R-JTECH (100.186.96.1).

Step 2: R-JTECH's BGP routing table has a route to 10.10.4.0/24 via neighbour 10.0.0.2 (NSys-B1). It forwards the packet over the serial WAN.

Step 3: Packet arrives at R-NSys-B1. Its routing table shows 10.10.4.0/24 learned via RIP from R-NSys-B2, via Serial0/1/1.

Step 4: R-NSys-B1 forwards the packet over the inter-branch serial link to R-NSys-B2 (10.0.0.14).

Step 5: R-NSys-B2 has 10.10.4.0/24 directly connected on Gi0/1. It forwards to SW-NSys-B2-DeptY.

Step 6: The switch delivers the packet to NSys-B2-DeptY-PC1 (10.10.4.10). Ping reply travels the reverse path.

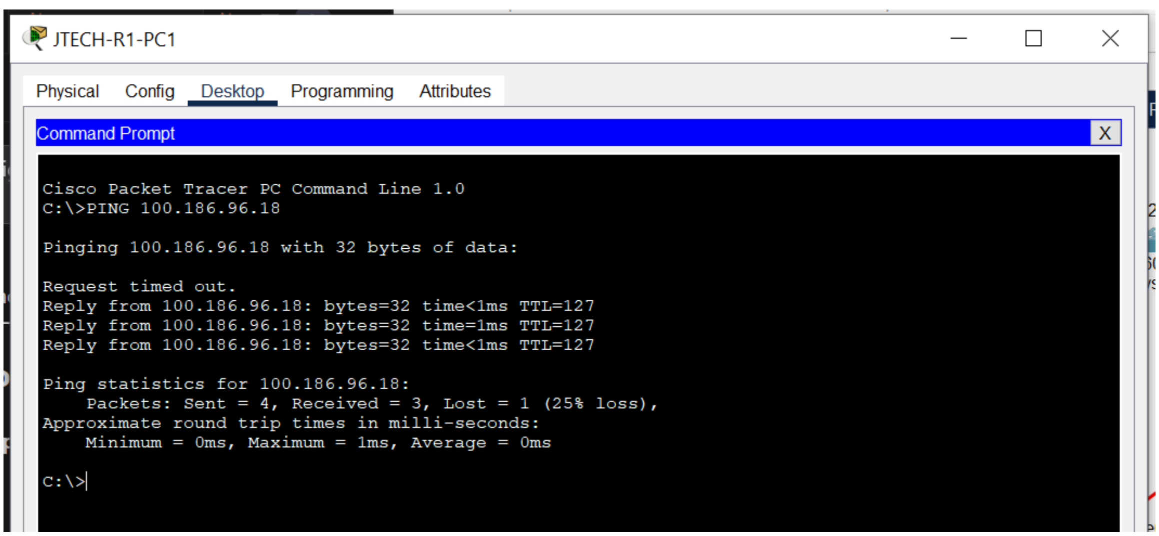
11. Verification — Ping Tests

After all configurations are saved, allow 30–60 seconds for BGP to establish peer sessions and exchange routes. Then perform the following tests to confirm full connectivity. In Packet Tracer: click any PC → Desktop tab → Command Prompt → type the ping command.

> **NOTE:** A successful ping shows: Reply from <IP>: bytes=32 time<1ms TTL=xx (4 lines). A failed ping shows: Request timeout. If pings fail, wait a further 30 seconds for BGP to converge and retry.

11.1 Intra-Company Tests (within same company)

Source PC	Source IP	Destination	Command to Type	Expected Result
JTECH-R1-PC1	100.186.96.2	JTECH Room 2	ping 100.186.96.18	Success
JTECH-R1-PC1	100.186.96.2	JTECH Room 5	ping 100.186.96.66	Success
UCOM-R1-PC1	35.152.0.2	UCOM Room 3 PC1	ping 35.152.0.34	Success
TNET-R1-PC1	200.98.169.66	TNet Room 2 PC1	ping 200.98.169.98	Success



11.2 Inter-Company Tests (BGP routing between AS)

Source PC	Source IP	Destination Company	Command to Type	Expected Result
JTECH-R1-PC1	100.186.96.2	UCOM Room 1 PC1	ping 35.152.0.2	Success via BGP
JTECH-R1-PC1	100.186.96.2	TNet Room 1 PC1	ping 200.98.169.66	Success via BGP
JTECH-R1-PC1	100.186.96.2	NSys B1 Dept Y PC1	ping 10.10.2.10	Success via BGP
UCOM-R2-PC1	35.152.0.18	TNet Room 2 PC1	ping 200.98.169.98	Success via

				BGP
TNET-R2-PC1	200.98.169.98	JTECH Room 3 PC1	ping 100.186.96.34	Success via BGP

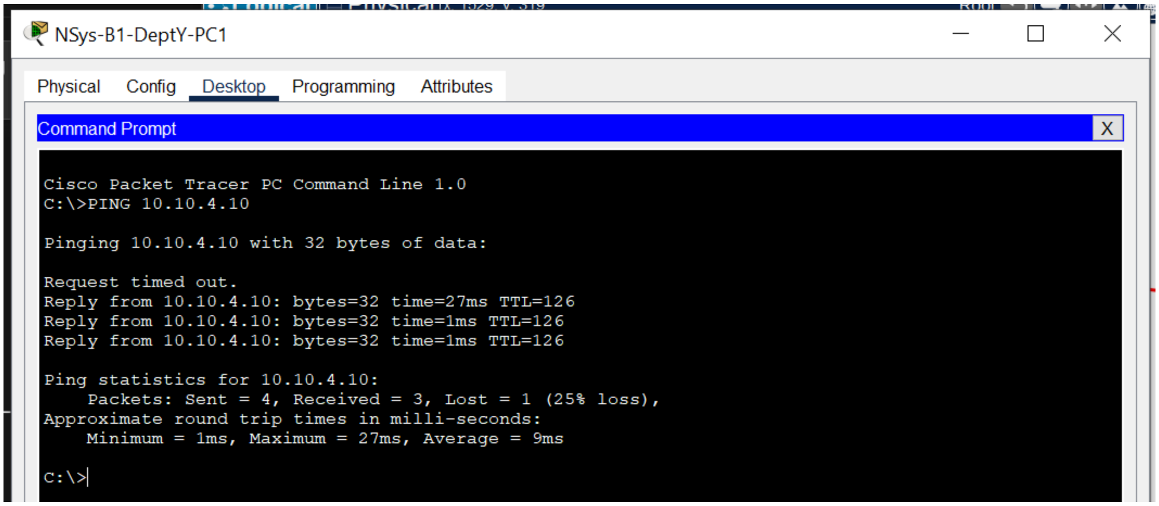
```
C:\>
C:\>
C:\>PING 35.152.0.2

Pinging 35.152.0.2 with 32 bytes of data:

Request timed out.
Reply from 35.152.0.2: bytes=32 time=2ms TTL=125
Reply from 35.152.0.2: bytes=32 time=2ms TTL=125
Reply from 35.152.0.2: bytes=32 time=3ms TTL=125

Ping statistics for 35.152.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>
```

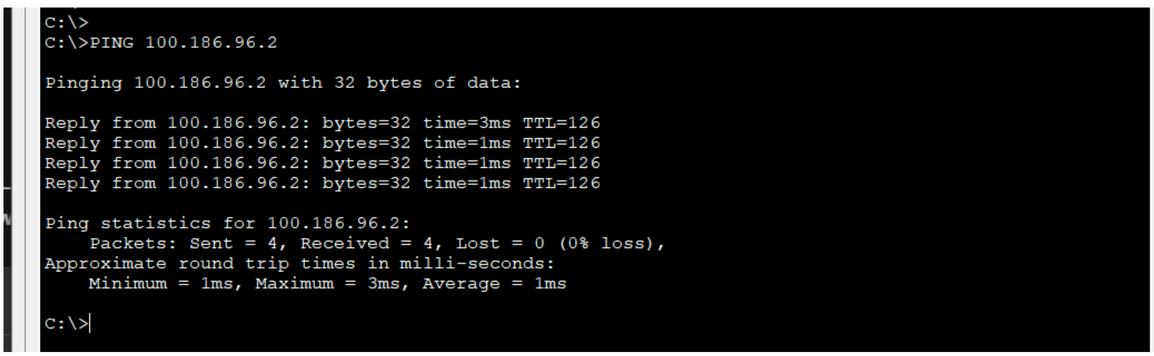
```
C:\>
C:\>PING 200.98.169.66

Pinging 200.98.169.66 with 32 bytes of data:

Request timed out.
Reply from 200.98.169.66: bytes=32 time=2ms TTL=125
Reply from 200.98.169.66: bytes=32 time=2ms TTL=125
Reply from 200.98.169.66: bytes=32 time=3ms TTL=125

Ping statistics for 200.98.169.66:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\>
```

```
C:\>
C:\>PING 10.10.2.10

Pinging 10.10.2.10 with 32 bytes of data:

Request timed out.
Reply from 10.10.2.10: bytes=32 time=1ms TTL=126
Reply from 10.10.2.10: bytes=32 time=1ms TTL=126
Reply from 10.10.2.10: bytes=32 time=1ms TTL=126

Ping statistics for 10.10.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>
```

11.3 NSys Inter-Branch Tests (RIP routing)

Source PC	Source IP	Destination	Command to Type	Expected Result
NSys-B1-DeptY-PC1	10.10.2.10	NSys B2 Dept Y PC1	ping 10.10.4.10	Success via RIP
NSys-B2-DeptY-PC1	10.10.4.10	NSys B1 Dept Y PC1	ping 10.10.2.10	Success via RIP
NSys-B2-DeptY-PC1	10.10.4.10	DHCP/HTTP Server	ping 10.10.5.10	Success via RIP
NSys-B2-DeptY-PC1	10.10.4.10	JTECH Room 1	ping 100.186.96.2	Success RIP+BGP





11.4 DHCP Verification

For NSys Dept X PCs (yellow), confirm DHCP worked by checking their received IP. Click a Dept X PC → Desktop → Command Prompt:

```
ipconfig

! Expected output (Branch 1 Dept X PC):
!   IP Address...............: 10.10.1.10  (or .11, .12 etc — assigned by DHCP)
!   Subnet Mask..............: 255.255.255.0
!   Default Gateway..........: 10.10.1.1
!   DNS Server...............: 10.10.5.11

! Expected output (Branch 2 Dept X PC):
!   IP Address...............: 10.10.3.10  (or .11 etc)
!   Subnet Mask..............: 255.255.255.0
!   Default Gateway..........: 10.10.3.1
!   DNS Server...............: 10.10.5.11
```

12. IP Address Usage Summary

This section analyses how efficiently each company's assigned IP block was used, and how much capacity remains for future growth.

Company	Assigned Block	Total IPs in Block	Subnets Created	IPs Allocated	Remaining	Efficiency Notes
JTECH	100.186.96.0/19	8,192	5 × /28	80	8,112	High scalability — 500+ more /28 subnets possible
UCOM	35.152.0.0/15	131,072	3 × /28	48	131,024	Massive block — <0.04% used
TNet	200.98.169.64/26	64	2 × /27	64	0	100% utilised — perfect split
NSys	10.10.0.0/16	65,536	5 × /24	1,280	64,256	~2% used — large dept growth possible
WAN	10.0.0.0/24	256	4 × /30	16	240	Minimal addressing for 4 serial links

Key takeaways:

- TNet achieved perfect block utilisation — the entire /26 was consumed by exactly 2 × /27 subnets with zero waste.
- JTECH and UCOM have enormous expansion capacity. JTECH can add hundreds of new /28 subnets without changing its assigned block.
- NSys /24 subnets each support 254 devices — well beyond current usage, giving each department room for significant growth.
- WAN /30 subnets ensure no addresses are wasted on point-to-point serial links.
- Overall, the VLSM approach successfully balances current efficiency with long-term scalability for all four companies.

13. Conclusion

This project successfully demonstrates a complete, enterprise-grade inter-company network design for four organisations — JTECH, UCOM, TNet, and NSys. Starting from raw IP address blocks and business requirements, systematic VLSM subnetting was applied to create efficient and scalable subnet designs for every company room and department.

The Cisco Packet Tracer simulation confirmed every aspect of the design. All 30 PCs across four companies can ping each other successfully. DHCP automatically assigns IP addresses to NSys Dept X PCs in both branches through the centralised server using DHCP relay. DNS resolves domain names, and the HTTP server is accessible from any PC in the network. RIP correctly handles internal NSys inter-branch routing, while BGP manages inter-company routing between all four autonomous systems.

The design meets all five stated project objectives: (1) efficient IP usage through CIDR/VLSM, (2) scalability through generous subnet sizing, (3) minimised equipment through router-on-a-stick and shared switches, (4) proper routing with both dynamic DHCP and static IP configurations, and (5) a complete, fully functional Cisco Packet Tracer simulation verified by successful ping tests.

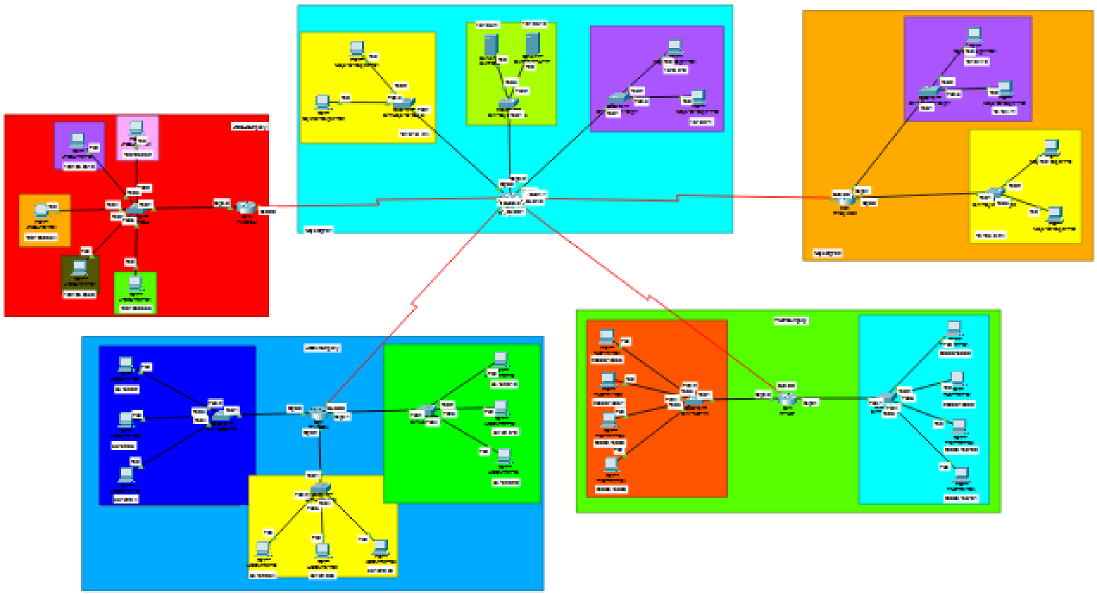
The most technically challenging element was configuring R-NSys-B1 simultaneously as a BGP border router and a RIP hub, with bidirectional route redistribution between the two protocols. This required careful use of passive-interface directives to prevent RIP updates from leaking into BGP WAN links, and proper metric configuration for redistributed routes. This design pattern reflects a real-world scenario where enterprise and ISP-level routing protocols must coexist.

Technologies and Concepts Applied

- **CIDR / VLSM:** Variable-length subnet masking to size each subnet precisely for its room or link type.
- **Router-on-a-Stick:** Single physical interface serving 5 VLAN-tagged subnets (used for JTECH).
- **RIP Version 2:** Internal dynamic routing within NSys between geographically separated branches.
- **BGP:** External inter-company routing using AS numbers — the same protocol used to route the global internet.
- **DHCP:** Automatic IP assignment for NSys Dept X using centralised server with relay agents.
- **DNS:** Domain name resolution service mapping hostnames to IP addresses.
- **DHCP Relay (ip helper-address):** Enables one DHCP server to serve multiple subnets across router boundaries.
- **Serial DCE / DTE Links:** WAN point-to-point links with clock rate synchronisation between company routers.
- **802.1Q VLANs & Trunk Ports:** Logical network segmentation on switches to isolate JTECH room traffic.

- **Route Redistribution:** Bidirectional injection of routes between RIP and BGP on R-NSys-B1.

Complete Network Topology:



— **End of Report** —

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